

Below is the Link to the google drive:

https://drive.google.com/drive/folders/1BngM7_9ilp9iultvt4wQdHqCWvu3tXXf?usp=sharing

Step-by-Step Instructions: Expanded Wine Market Analysis

1. Environment Setup

Ensure you have Python installed (3.8+ recommended). Install the required libraries using the following command in your terminal:

```
pip install pandas matplotlib seaborn nbformat wordcloud
```

2. Project Phase Scripts

The following scripts represent the stages of the investigation:

- `clean_data.py`: Preprocessing & Dataset Merger
- `analysis_correlation.py`: Price vs. Quality Analysis
- `analysis_market.py`: Market Share & Geography
- `analysis_tasters.py`: Critic Score Consistency
- `analysis_price_segments.py`: Price Tiering (Budget to Luxury)
- `analysis_vintage_extraction.py`: Year Data Extraction
- `analysis_heatmap_geo.py`: Regional Specialization Heatmap
- `analysis_wordcloud_high.py`: High-Rating Descriptor Profile
- `analysis_wordcloud_low.py`: Low-Rating Descriptor Profile
- `analysis_top_value_bottles.py`: Points-per-Dollar ROI Study
- `analysis_winery_stars.py`: Identifying Leading Producers
- `generate_final_expanded_notebook.py`: Final Report Consolidation

3. Investigation Flow (12 Terminal Interactions)

Run these scripts in sequence to replicate the full Investigation:

1. python clean_data.py (Initial Data Prep)
2. python analysis_correlation.py (Initial Correlations)
3. python analysis_market.py (Variety & Country Volume)
4. python analysis_tasters.py (Taster Bias Check)
5. python analysis_price_segments.py (Market Tiering)
6. python analysis_vintage_extraction.py (Vintage Analysis)
7. python analysis_heatmap_geo.py (Geo-Variety Mapping)
8. python analysis_wordcloud_high.py (Linguistic profiling - High)
9. python analysis_wordcloud_low.py (Linguistic profiling - Low)
10. python analysis_top_value_bottles.py (Value Opportunity List)
11. python analysis_winery_stars.py (Star Producer Identification)
12. python generate_final_expanded_notebook.py (Creates MiraclePython.ipynb)

4. Final Deliverables

Upon completion, you will find the comprehensive analysis and assortment strategy in:

MiraclePython.ipynb (Open with Jupyter or VS Code)